

Course Introduction and Administration

COMPSCI 753: Algorithms for Massive Data

Instructors: Miao Qiao, Zijian Huang and Ruofan Wang



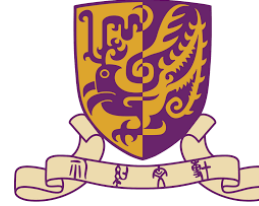
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Course description

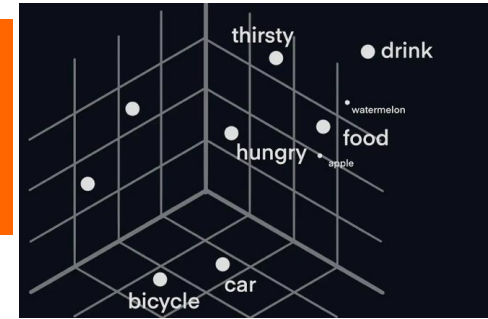
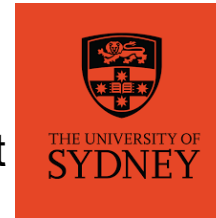
- Modern applications such as e-commerce, social networks, and location-based services are expecting efficient big data solutions.
- **Course aims:**
 - Expose practitioners to computational challenges in managing and mining big data.
 - Introduce a wide spectrum of advanced algorithms that underpin big data analytics and knowledge discovery.
 - Explore the best-practices and the current developments of big data research.

COMPSCI 753 Instructors

- Dr. Miao Qiao
 - Home page: <https://miaoqiao.github.io/>
 - Email: miao.qiao@auckland.ac.nz
 - Office: Building 303, Room 524
 - Office hours: Email for a Zoom appointment

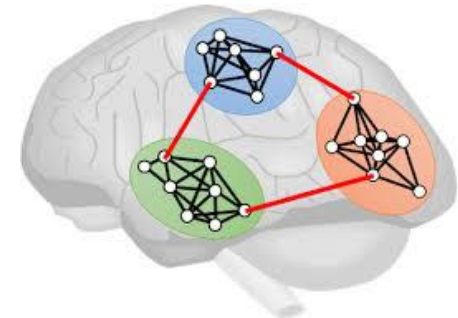
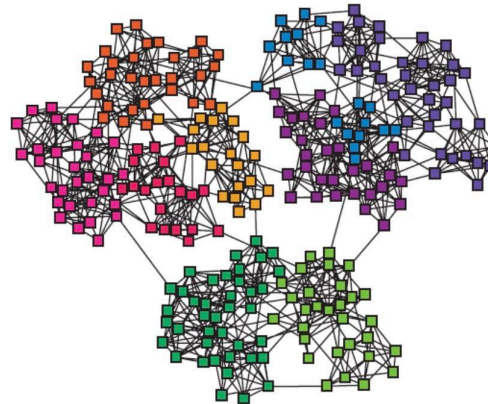


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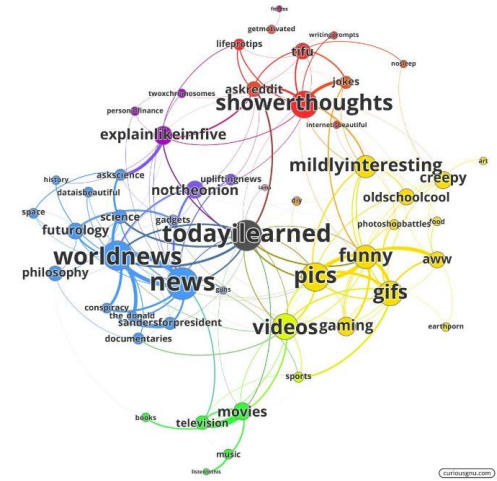
- **Education:** Chinese University of Hong Kong, Hong Kong SAR

- **Current research activities:**
 - Vector data management
 - Graph search and analytics
 - Brain network analytics



COMPSCI 753 Instructors

- Zijian Huang
 - Home page: <https://nelsonhuangzijian.github.io/>
 - Email: zijian.huang@auckland.ac.nz
 - Office: Newmarket Campus, Building 903, Room 435
 - Office hours: Email for a Zoom appointment
- **Education:** PhD candidate at The University of Auckland
- **Current research activities:**
 - Knowledge Representation Learning and Reasoning
 - Knowledge Graphs Embedding
 - Complex Logical Query Answering
 - Natural Language Processing
 - Document Summarization

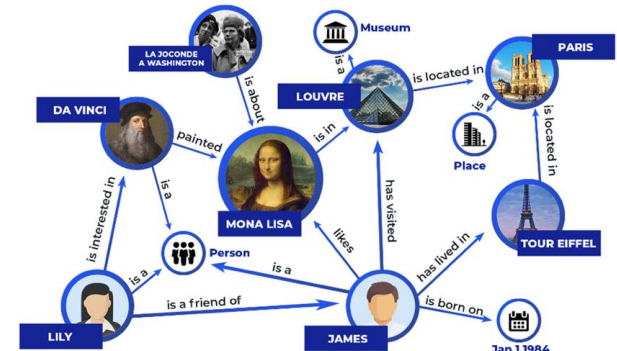


COMPSCI 753 Instructors

- Ruofan Wang
 - Home page: <https://profiles.auckland.ac.nz/ruofan-wang>
 - Email: ruofan.wang@auckland.ac.nz
 - Office: Newmarket Campus, Building 903, Room 435
 - Office hours: 10 am–11 am, Thursday (Week 1 to Week 3)
 - Zoom Link: [Canvas](#)

- **Education:** PhD candidate at The University of Auckland

- **Current research activities:**
 - Multimodal Representation Learning
 - Knowledge Graphs
 - Multimodal Reasoning



Big data analytics

- The Moore's law of big data:
 - “We have generated more than 90% of data in the last two years itself. ”
- Big data analytics:
 - “How to turn big data into value? ”



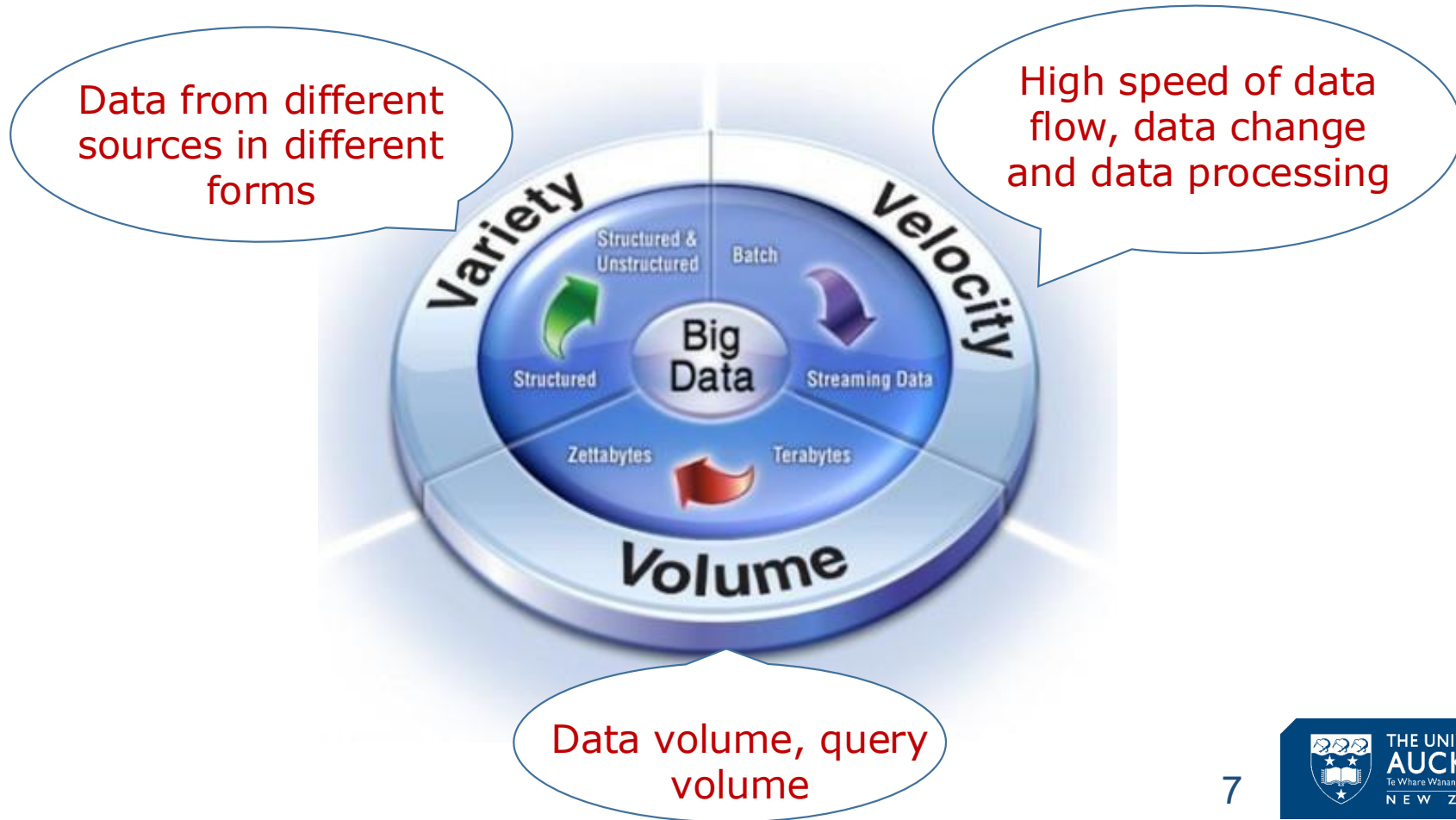
Created by: eDiscovery Today & LTMG

Big data and the 3 V's properties

- "Big data represents the information assets characterized by such a **high volume**, **velocity** and **variety** to require specific technology and analytical methods for its transformation into value."

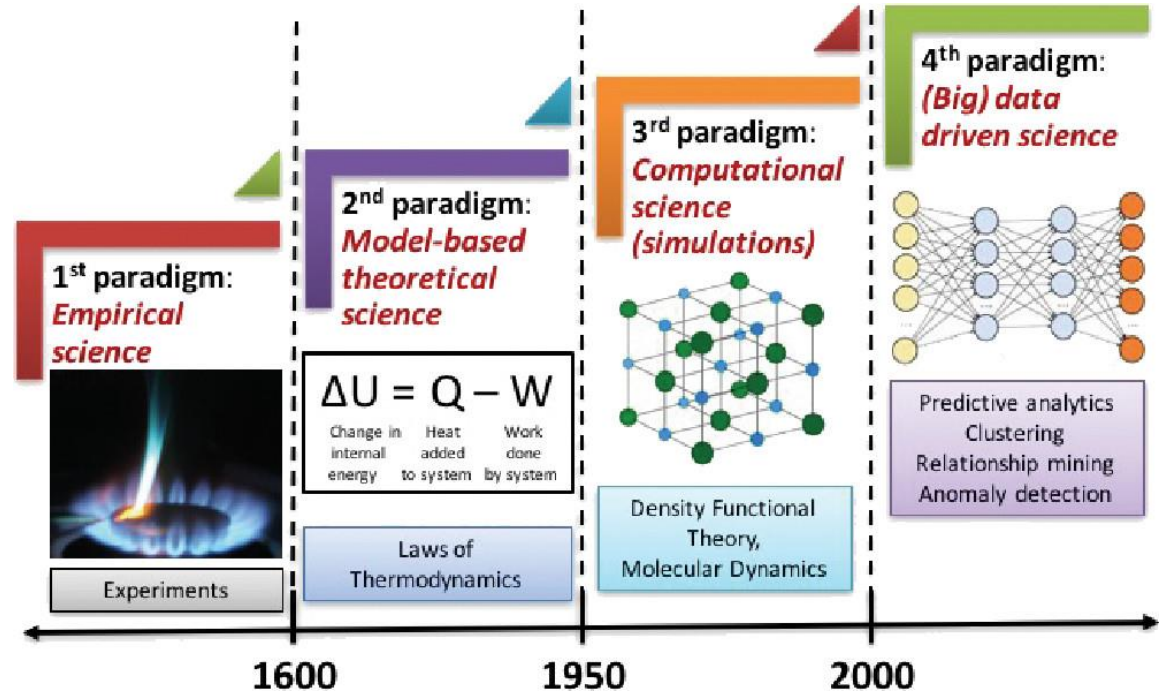
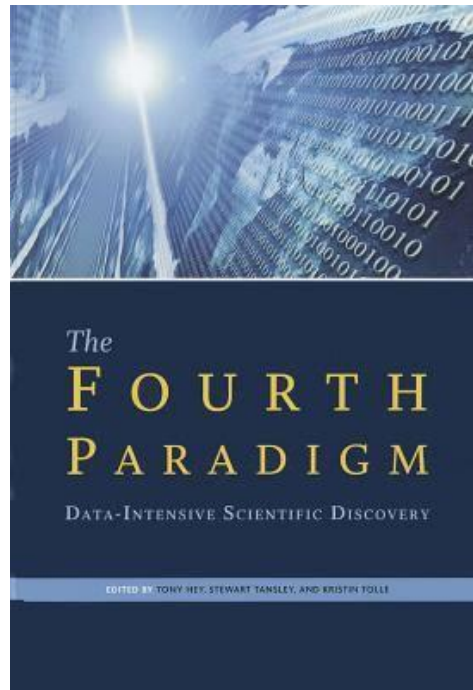


Challenges of big data analytics



Data Science

- “The Four Paradigm” by Tony Hey



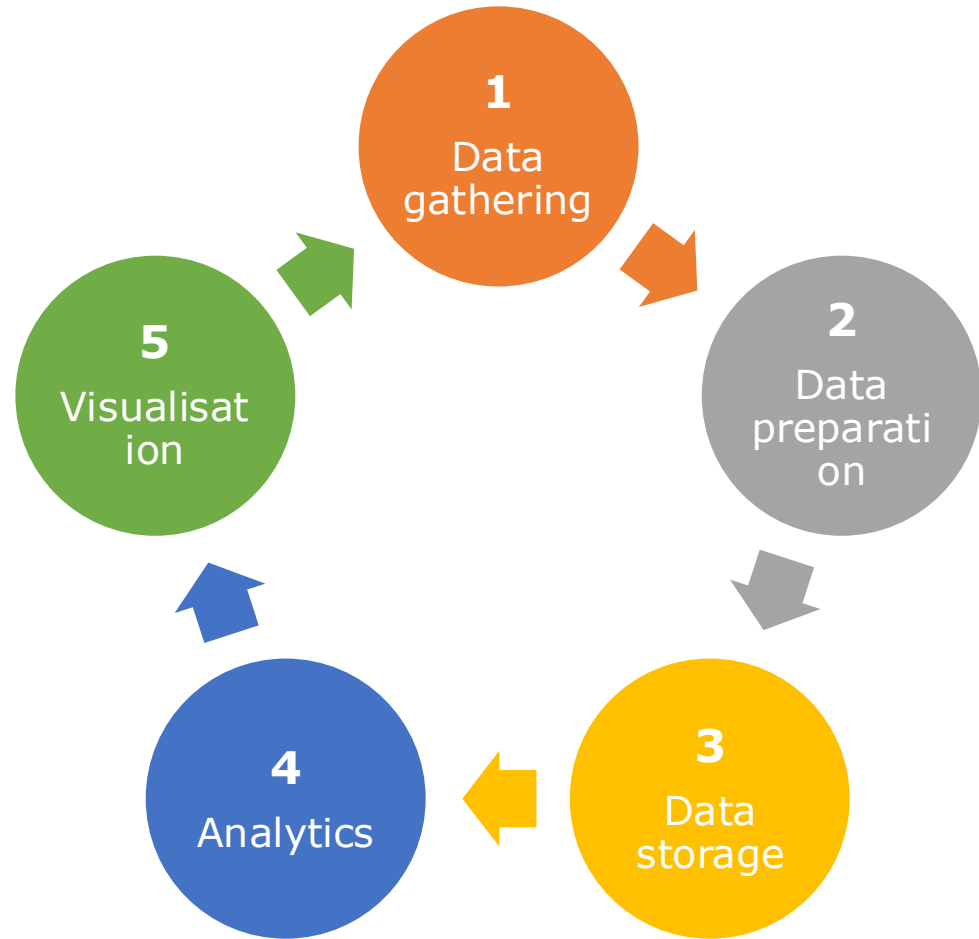
Source: Ankit Agrawal; Alok Choudhary; *APL Materials* 4, 053208 (2016)

Data Science

The data needs to be

- Stored
- Managed
- **Analysed (We will cover this)**

Data Mining
≈ **Big Data**
≈ **Predictive Analytics**
≈ **Data Science**



Data Science Life Cycle

Data science jobs in NZ

Data Analyst

Kaitātari Raraunga

Alternative titles for this job

Data analysts identify and describe data trends using statistics and specialised software to help organisations achieve their business aims.

Pay

Data analysts usually earn

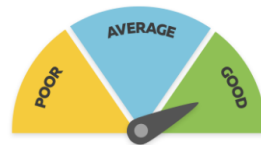
\$90K-\$120K per year

Data scientists usually earn

\$110K-\$170K per year

Source: Absolute IT, 2023.

Job opportunities



Chances of getting a job as a data analyst are good due to a shortage of workers.

Length of training



2-3 years of training usually required.

Source:

<https://www.careers.govt.nz>

Data Mining

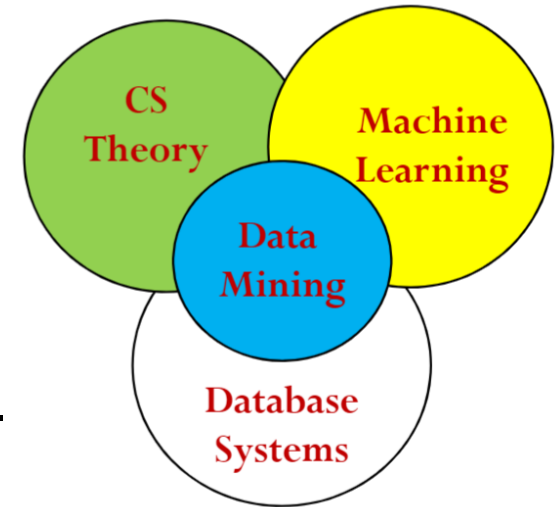
- Discover patterns and models from the big data!
- **Descriptive tasks**
 - Find human-interpretable patterns that describe the data
 - **Example:** Clustering, anomaly detection
- **Predictive tasks**
 - Use some variables to predict unknown or future values of other variables
 - **Example:** Recommender systems, regressions...

发现已有模式

预测未来模式

Data Mining: Cultures

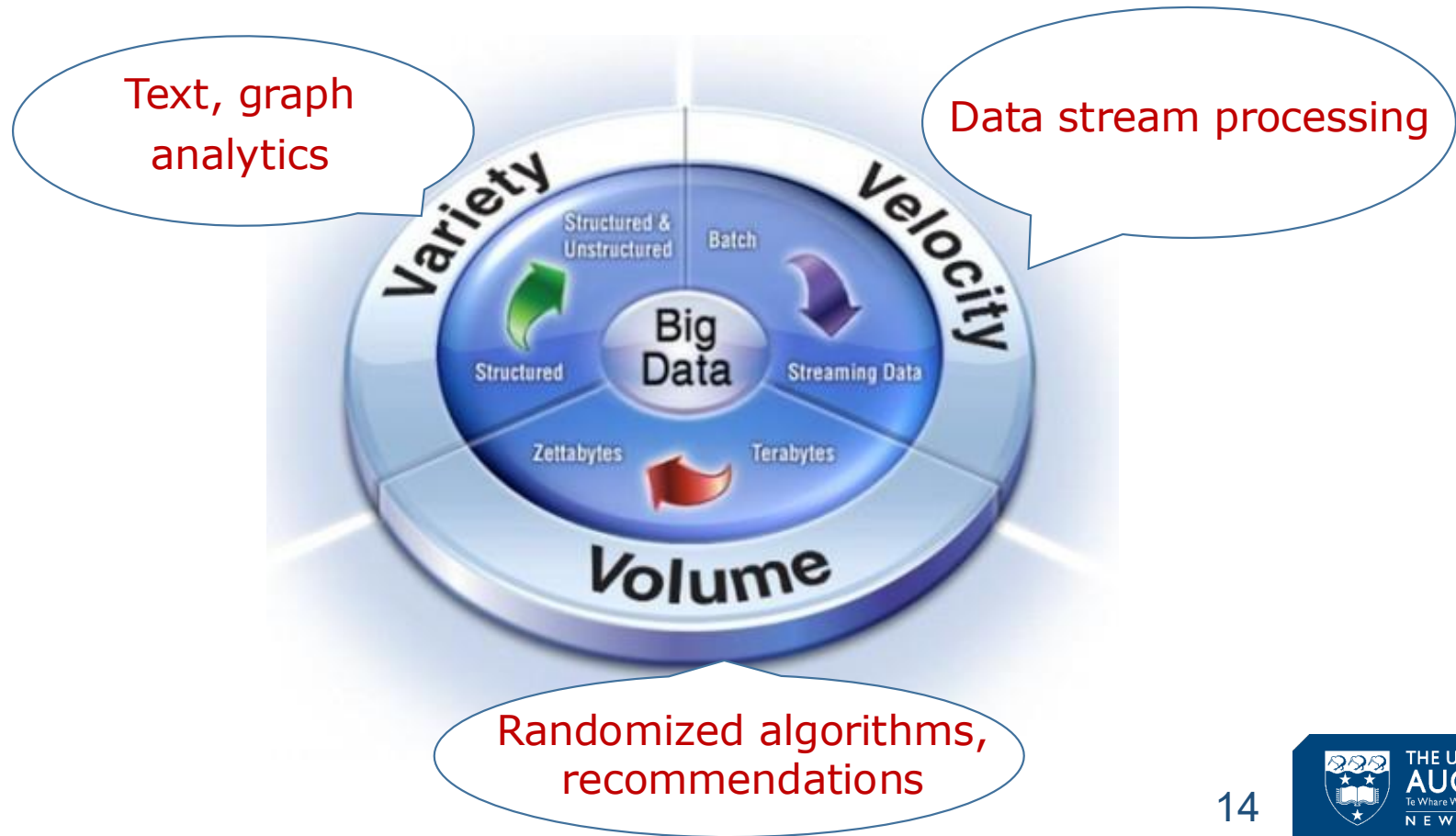
- Data mining overlaps with:
 - **Databases**: Large-scale data management
 - **Machine learning**: Large-scale complex models
 - **CS Theory**: Efficient (randomized) algorithms
- This paper: COMPSCI-753
 - We cover all these overlaps but more stress on **scalable algorithms to handle the 3V's of big data**.
 - Focus on computational and statistical challenges of big data.



What will you learn?

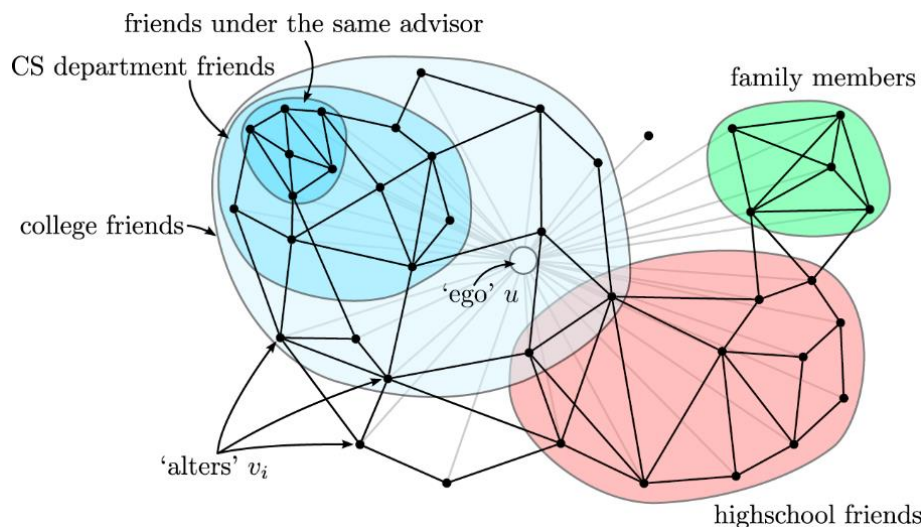
- Analyse different types of data:
 - Data is high-dimensional such as text
 - Data is graph
 - Data is infinite/never-ending
- Solve real-world problems:
 - Similarity search
 - Web analytics
 - Social networks
 - Recommender systems

Course content



Course content

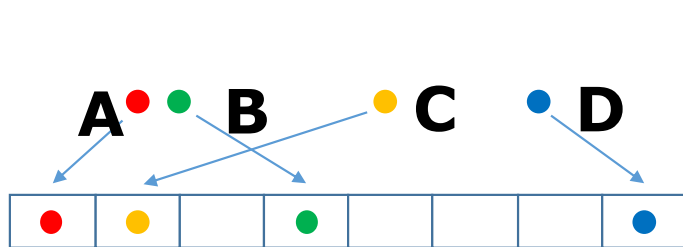
- **Part 1: Graph mining** (weeks 1-3 by Ruofan)
 - Challenges in mining big graphs, e.g., web, social networks
 - Techniques for efficient web search
 - Algorithms for mining social networks



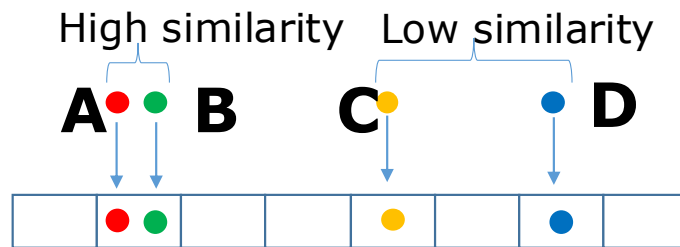
[McAuley, Leskovec: Discovering social circles in ego networks, 2012]

Course content

- **Part 2: Locality-sensitive hashing** (weeks 3-6 by Zijian)
 - Basic probability and universal hashing
 - Similarity search applications
 - MinHash/SimHash for common Jaccard/cosine similarity in information retrieval
 - Locality-sensitive hashing for near duplicate detection and similarity search



General hashing



Locality-sensitive hashing

Course content

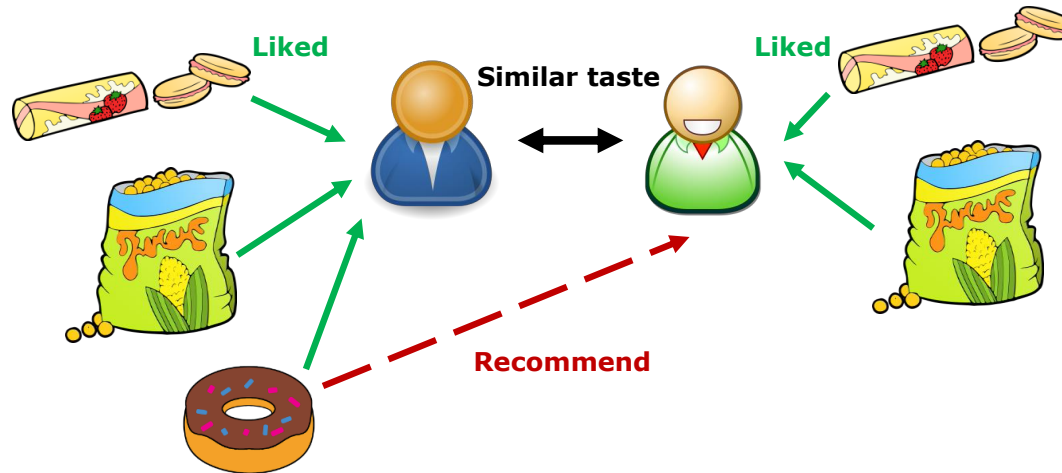
- **Part 3: Data stream** (weeks 7-9 by Miao)
 - Challenges on processing data stream
 - Randomized techniques for processing data stream: sampling, filtering, finding heavy-hitters, counting distinct, sketching
 - Accelerating machine learning with randomized techniques



Source: <https://blog.fastforwardlabs.com/2016/11/23/probabilistic-data-structure-showdown-cuckoo.html>

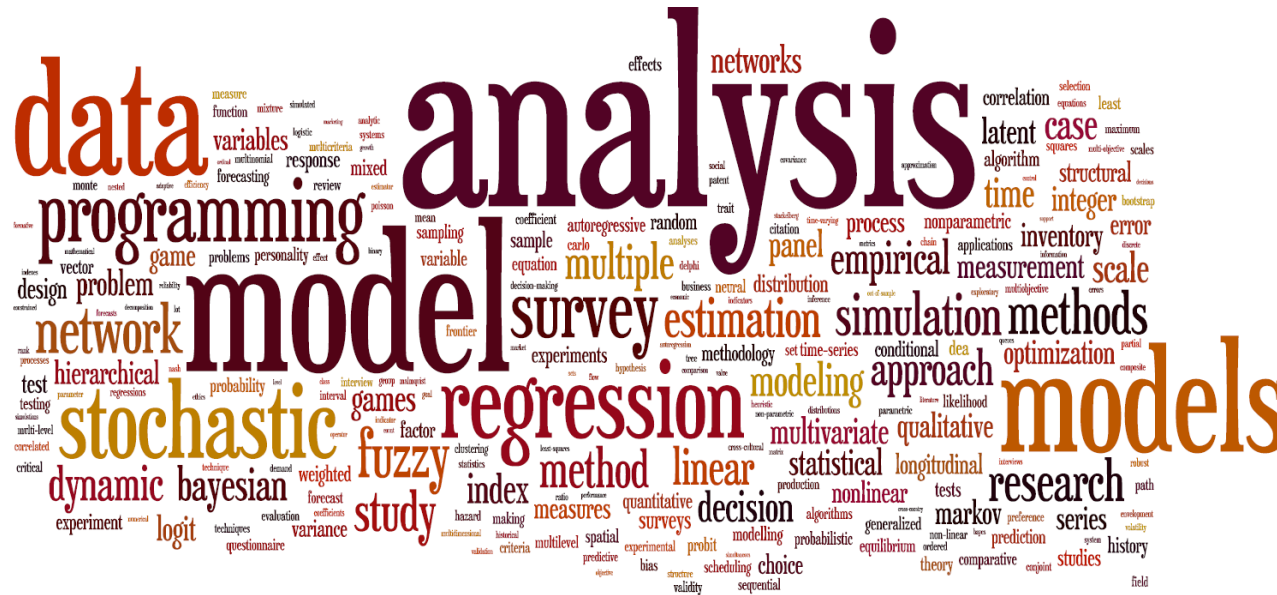
Course content

- **Part 4: Recommender systems** (weeks 9-10 by Miao)
 - Search v.s. recommendation
 - Recommendation algorithms: content-based approaches, memory-based collaborative filtering, model-based collaborative filtering
 - Advanced topics in recommender systems: context-aware recommendation, session-based recommendation



Course content

- **Part 5: Research presentations** (weeks 11, 12 by you)
 - Each group of students will present a research topic related to computational and statistical challenges of big data.



Learning outcomes & expectation

- Understand **computational bottlenecks** of processing, managing and mining big data.
- Understand **algorithmic tools** to handle big data analytics, including hashing, sampling and sketching.
- Design and analyze **efficient algorithms** for real-world big data analytics with variety of data types, including text, graph, stream.
- **Present technical ideas** and results in both verbal and written forms.
- The course provides a starting point for you to step in the research area. As a postgraduate student, you should be able to search for relevant materials to expand your knowledge
 - See: <https://canvas.auckland.ac.nz/courses/60599/files/7067446?wrap=1>

Learning resources

- **CANVAS:**
 - Slides and lecture recordings.
 - Textbooks, reading materials, references, research topics.
- Do strategic exercises and check model answers.
- Do textbook exercises and think of your own examples.
- Discuss with peers.
- Research articles available through the library.

Assessment overview

- **Must pass theory and practical**
- **Practical: 4 individual assignments (20%)**
 - **Programming** assignment (prefer Python)
 - Each assignment is worth 5% of your final mark
- **Theory: research project in group (30%)**
 - Presentation (15%)
 - Research report (15%)
- **Theory: Final exam (50%)**
 - 2 hrs Closed-book Paper-based Exam

Assessment details: Assignments

- **4 programming assignments:**

- **Submission:**

- A source code (prefer Python) with clear comments.
 - A **pdf file** reports the requested answers and your explanation for each task.

- **Assessment:**

- The marker will check your pdf reports and run the codes and compare with the model answers. If part of the results match the model answers, you will get partial mark. Details could be found in the assignment description.

- **We will check for plagiarism:**

- <https://theory.stanford.edu/~aiken/moss/>
 - **0 mark** for the detected cases
 - Read the Academic Integrity document on Canvas carefully!

Assignments due dates

- **Assignments:**
 - 1st due: 23:59 NZST, Sunday 17 August
 - 2nd due: 23:59 NZST, Sunday 14 September
 - 3rd due: 23:59 NZST, Sunday 5 October
 - 4th due: 23:59 NZST, Sunday 26 October
 - **You will have three weeks to work on each assignment!**
- **Late submissions:**
 - From **1** minute to **1** day late: **25%** penalty
 - From **1** day to **2** days late: **50%** penalty
 - More than **2** days late: **100%** penalty
- Let us know as early as possible (preferably three days prior to the deadline) for any reason you can't submit in time.

Assessment details: Research project

■ Research project:

- We provide a list of research topics and its reference about computational and statistical challenges of real-world big data applications.
- Each **team of up to 4 students** sends all of us **5 research topics** with priority before **5pm, Friday 1 August**. We use the **first-come-first-serve** principle to allocate your team to the topic.
- If you can't find a teammate: send us the top 5 choices of topics, we will find your teammate 😊

■ Components:

- Oral presentation (**15%**)
- Research report (**15%**)

Assessment details: Research project

■ Research Topics

Locality Sensitive Hashing

Entity Resolution
Fairness
Model Optimisation
Data Hashing
Clustering
Anti-abuse Detection

Stream

Density Estimation
Evolving Graph
Classification
Tensor Stream
Multi-Aspect Stream
User Profiling

Graph

Community Search
Community Detection
Clustering
Link Prediction
Personalized PageRank
Graph Representation

Recommender System

Aspect-based RS
Context-Aware RS
Deep Learning
Graph-based RS
Session-based RS
Topic model

Assessment details: Research project

- **Presentation (15%):**
 - **12'** presentation, time is strict (**12%**).
 - **3'** Q&A (**3%**).
- **Research report (15%):**
 - Literature review (**12%**) of at least **5 research articles** (including the seed paper) on your research topic.
 - Your new ideas or justification (**3%**).
 - Format: **6-8 pages** of IEEE proceeding format (double columns).
<https://www.ieee.org/conferences/publishing/templates.html>

Research project due dates

- **Research report:**

- Report abstract and structure: **Due by 23:59 NZST, Sunday, 14 September.** This abstract and structure **is not graded**, but we can give you some feedback if you submit your abstract before the deadline
- Final report: **Due by 23:59 NZST, Sunday, 2 November.**

- **Late submissions:**

- From **1** minute to **1** day late: **25%** penalty
- From **1** day to **2** days late: **50%** penalty
- More than **2** days late: **100%** penalty

- **Marking process:**

- Rubrics and marking schemes for both research presentation and report are on CANVAS.

Plagiarism

- **Plagiarism defined:**

- *“Using the work of other scholars or students when preparing coursework or writing an examination and pretending it is your own by not acknowledging where it came from.”*

- **Seeking advice:**

- Course coordinators, lecturers or tutors are the appropriate people with whom you should discuss how to appropriately use and acknowledge the work of others.

- **Consequences:**

- **Cheating is viewed as a serious offence by The University of Auckland.** Penalties are administered by the Discipline Committee of the Senate, and may include suspension or expulsion from the University.

How to avoid plagiarism

- Always do individual assignments by yourself.
- Never loan your code to another person.
- Never get code from a tutors. Several tutors have been caught giving the same code to all their students.
- Always reference the source for text you copy as part of the answer to an assignment.

Timetable

- **Lectures** (only **3** hours will be used per week, unless announced otherwise)
- **Tutorials** (announced beforehand and usually after finishing each topic)
- Tuesday, 11am-1pm, Clock Tower, Room 032
- Friday, 1pm-3pm, Clock Tower, Room 029

Timeline

- **Week 1-2:** Choose your teammate and research topic
- **Week 2:** Assignment 1 release
- **Week 4:** Assignment 1, **due** by **23:59 NZST, Sunday, 17 Aug, 2025**
- **Week 5:** Assignment 2 release
- **2-week break:**
 - Assignment 2, **due** by **23:59 NZST, Sunday, 14 Sep, 2025**
 - Assignment 3 release
 - Report abstract and structure, **due** by **23:59 NZST, Sunday, 14 Sep, 2025**
- **Week 8:**
 - Assignment 3, **due** by **23:59 NZST, Sunday, 5 Oct, 2025**
- **Week 9:** Assignment 4 release
- **Week 11-12:** Student presentations
- **Week 11:** Assignment 4 **due** by **23:59 NZST, Sunday, 26 Oct, 2025**
- **Week 12:** Research report, **due** by **23:59 NZST, Sunday, 2 Nov, 2025**